

using satellite data for applications such as earth observation, communication, and navigation.

## UNTAPPED POTENTIAL

India's tryst with space is not without its share of challenges. Compared with the Top 2 spacefaring powers, the US and China, the country stands at No. 7 in terms of satellites operated. It continues to depend on the West for imports of critical components for launch vehicles, spacecraft, and satellites. Furthermore, it has virtually no presence in activities such as space manufacturing, human space transport, space tourism, and high-altitude platforms and balloons. This should not come as a surprise, as the country's space budget is merely 0.05 per cent of its GDP. In comparison, the US spends 0.25 per cent of its GDP on space.

Per a study by Arthur D. Little, the country's space industry is expected to grow to \$40 billion by 2040, at a CAGR of 9.2 per cent, to reach a 4 per cent market share globally. But it says India can do better and touch \$100 billion by that year to corner 10 per cent market share.

Besides, the ongoing realignment in geopolitics has resulted in India being perceived as a counter to China. A surprise agreement inked during Prime Minister Narendra Modi's state visit to the US was India becoming the 27th signatory to the Artemis Accords, a multilateral agreement between the US and other countries to return humans to the moon by 2025. More recently, ahead of the G20 Summit in New Delhi, the US reiterated its commitment to send the first Indian astronaut to the International Space Station later this year.

"As a global space community, we have been trying to formulate



**MANECK BEHRAMKAMDIN**  
ASSOCIATE VP & BUSINESS HEAD,  
GODREJ AEROSPACE

**"The opening up of the space sector to private players and start-ups is a significant step towards augmenting the competitiveness of the Indian space programme"**



**CHIRAG DOSHI**  
MANAGING DIRECTOR & CEO,  
WALCHANDNAGAR INDUSTRIES

**"Since our division's humble beginnings in the early 1970s, we boast of 40 per cent of our resources, both men and machines, exclusively working [on] India's space programme"**

some shared perspectives on space exploration, how it should be accomplished, and who should benefit. The Artemis Accords were one of the first recent efforts to advance [those] original principles laid out almost 50 years ago," Elizebeth Varghese, People in Space-Leader at Deloitte and author of *Stellar Sin-*

*gularity: Navigating the Spacefaring Economy*, tells BT.

Neha Singh, Chairperson and MD of data services provider Tracxn, feels this expanding India-US partnership will especially give a fillip to start-up funding. The ecosystem received \$112 million in 2022, which is 17 per cent higher than in 2021, according to Tracxn.

So, how soon can India expect to see its own Elon Musk and SpaceX? Arthur D. Little's Singh feels India still has a lot of ground to cover in terms of developing its R&D base and facilitating options for raising easy financing. "At the same time, we must not forget that Musk has taken years to reach where he is today. And that wouldn't have happened without policy support from the US government. In the last couple of years, India's space programme has moved into the next orbit. Provided this continues, we have the potential to produce our own versions of Musk and SpaceX," suggests Singh.

Meanwhile, Deloitte's Ananthasayanam has high expectations from spacetech start-ups. "It's just a matter of finding the right sponsor or person with the right financial muscle to back them up...Some of these start-ups are working in areas such as Space Situational Awareness (SSA), green and electric propulsion and laser-based communications that only a very few companies are into, globally." SSA is the knowledge of the space environment, including the location, orbit, and function of man-made objects as well as natural objects such as asteroids and comets.

If things remain on track, a thousand SpaceXs could bloom in India. **BT**

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